tasks W-02

```
import numpy as np
import pandas as pd
pd.options.mode.chained_assignment = None
import pickle

import os
from importlib import reload

import matplotlib.pyplot as plt
import matplotlib
# import seaborn as sns
%matplotlib inline
plt.style.use('ggplot')
from matplotlib.ticker import MultipleLocator
```

load data

```
df_comp_details = pd.read_csv('duopoly_competition_details.csv')
df_comp_details.fillna(0, inplace=True)
df_comp_details.head(3)
```

```
  competition_id  selling_season  selling_period   competitor_id  \
0         3HmCSH               1               1  SaffronDolphin
1         3HmCSH               1               2  SaffronDolphin
2         3HmCSH               1               3  SaffronDolphin

   price_competitor  price  demand  competitor_has_capacity  \
0              56.1   66.4       0                     True
1              66.4   66.4       0                     True
2              66.4   66.4       0                     True

   calculation_duration  errors
0                 0.005     0.0
1                 0.006     0.0
2                 0.007     0.0
```

(1) find based on your own competition details CSV the best (max revenue) capacity utilization curve

(a) compute the revenue for all selling_seasons and competitions

```
df_comp_details['revenue'] =  """ADD CODE HERE"""
df_comp_details.tail(5)
```

```
      competition_id  selling_season  selling_period   competitor_id  \
39995         rPaVZC             100              96  AttentiveOriole
39996         rPaVZC             100              97  AttentiveOriole
39997         rPaVZC             100              98  AttentiveOriole
39998         rPaVZC             100              99  AttentiveOriole
39999         rPaVZC             100             100  AttentiveOriole

       price_competitor  price  demand  competitor_has_capacity  \
39995              13.5   13.7       3                    False
39996              13.7   13.7       0                    False
39997              13.7   13.7       0                    False
39998              13.7   13.7       0                    False
39999              13.7   13.7       0                    False

       calculation_duration  errors  revenue
39995                 0.004     0.0     41.1
39996                 0.004     0.0      0.0
39997                 0.004     0.0      0.0
39998                 0.004     0.0      0.0
39999                 0.005     0.0      0.0
```

```
df_comp_details['unique_selling_season_key'] = """ADD CODE HERE"""
df_comp_details.head(2)
```

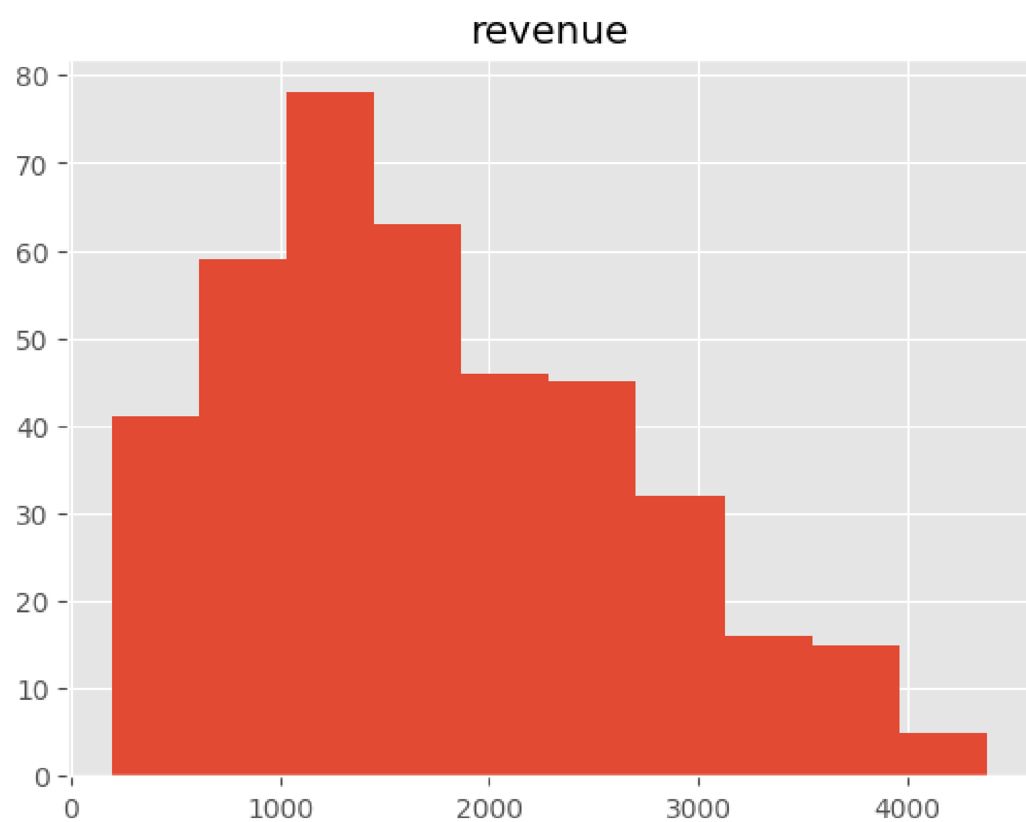
```
  competition_id  selling_season  selling_period   competitor_id  \
0         3HmCSH               1               1  SaffronDolphin
1         3HmCSH               1               2  SaffronDolphin

   price_competitor  price  demand  competitor_has_capacity  \
0              56.1   66.4       0                     True
1              66.4   66.4       0                     True

   calculation_duration  errors  revenue unique_selling_season_key
0                 0.005     0.0      0.0                  3HmCSH_1
1                 0.006     0.0      0.0                  3HmCSH_1
```

```
dfx_rev = df_comp_details.groupby('unique_selling_season_key').agg({
    'revenue' : 'sum'
}).reset_index()
dfx_rev
```

```
    unique_selling_season_key  revenue
0                    3HmCSH_1    363.7
1                   3HmCSH_10   1266.4
2                  3HmCSH_100    399.7
3                   3HmCSH_11   1652.7
4                   3HmCSH_12   1572.3
..                        ...      ...
395                 rPaVZC_95   2844.8
396                 rPaVZC_96   1784.6
397                 rPaVZC_97   4376.1
398                 rPaVZC_98   2629.0
399                 rPaVZC_99   2842.1

[400 rows x 2 columns]
```

```
dfx_rev.hist()
```

```
array([[<Axes: title={'center': 'revenue'}>]], dtype=object)
```



```
dfx_rev.describe()
```

```
             revenue
count   400.000000
mean   1740.731250
std     939.469504
min     196.900000
25%    1034.900000
50%    1570.350000
75%    2401.250000
max    4376.100000
```

get the selling season keys with the top 20% revenue generation

```
dfx_rev.quantile(0.8)
```

```
/tmp/ipykernel_705/3896314509.py:1: FutureWarning: The default value
of numeric_only in DataFrame.quantile is deprecated. In a future
version, it will default to False. Select only valid columns or
specify the value of numeric_only to silence this warning.
  dfx_rev.quantile(0.8)
```

```
revenue    2573.22
Name: 0.8, dtype: float64
```

```
top_selling_seasons = dfx_rev[    """ADD CODE
HERE"""   ].unique_selling_season_key.unique()
top_selling_seasons[:10]
```

```
array(['3UDGYU_57', '3UDGYU_96', 'bh2y6f_1', 'bh2y6f_11', 'bh2y6f_14',
       'bh2y6f_15', 'bh2y6f_19', 'bh2y6f_21', 'bh2y6f_22',
'bh2y6f_23'],
      dtype=object)
```

```
# quick double check
dfx_rev[dfx_rev.unique_selling_season_key.isin(top_selling_seasons)].r
evenue.mean()
```

```
3188.67
```

(b) compute the capacity utilization curves per selling season

```
dfp_util.tail(15)
```

unique_selling_season_key	3HmCSH_1	3HmCSH_10	3HmCSH_100	3HmCSH_11	3HmCSH_12	3HmCSH_13	3HmCSH_14	3HmCSH_15	3HmCSH_1
selling_period									
86	0.38	0.37	0.38	0.40	0.39	0.40	0.37	0.32	0.3
87	0.39	0.38	0.39	0.41	0.40	0.41	0.38	0.33	0.3
88	0.40	0.39	0.41	0.42	0.41	0.42	0.39	0.34	0.3
89	0.41	0.40	0.42	0.44	0.42	0.43	0.41	0.35	0.3
90	0.43	0.42	0.43	0.45	0.43	0.44	0.42	0.36	0.3
91	0.44	0.43	0.44	0.46	0.44	0.45	0.43	0.37	0.3
92	0.45	0.44	0.45	0.47	0.45	0.46	0.44	0.38	0.3
93	0.46	0.45	0.47	0.48	0.47	0.47	0.46	0.39	0.4
94	0.48	0.46	0.48	0.49	0.48	0.48	0.47	0.39	0.4
95	0.49	0.47	0.49	0.51	0.49	0.49	0.48	0.40	0.4
96	0.50	0.49	0.50	0.52	0.50	0.51	0.49	0.41	0.4
97	0.51	0.50	0.52	0.53	0.52	0.52	0.51	0.42	0.4
98	0.53	0.51	0.53	0.54	0.53	0.53	0.52	0.43	0.4
99	0.54	0.52	0.54	0.56	0.54	0.54	0.53	0.44	0.4
100	0.55	0.54	0.55	0.57	0.55	0.55	0.54	0.45	0.4

15 rows × 400 columns

```
dfp_util = pd.pivot_table(df_comp_details,
                          """ADD CODE HERE"""
)

dfp_util.tail()

unique_selling_season_key  3HmCSH_1  3HmCSH_10  3HmCSH_100  3HmCSH_11  \
selling_period

96                                0          0           0          3

97                                0          2           0          1

98                                0          0           0          0

99                                0          0           0          0

100                               0          1           0          0


unique_selling_season_key  3HmCSH_12  3HmCSH_13  3HmCSH_14  3HmCSH_15  \
selling_period

96                                 0          0          0          1
```

```
97                                 0          1          0          1

98                                 0          4          0          2

99                                 0          0          0          0

100                                0          2          0          1


unique_selling_season_key  3HmCSH_16  3HmCSH_17  ...  rPaVZC_90
rPaVZC_91  \
selling_period                                   ...

96                                 3          3  ...          2
0
97                                 3          1  ...          1
2
98                                 2          0  ...          1
2
99                                 1          0  ...          1
2
100                                1          0  ...          0
0

unique_selling_season_key  rPaVZC_92  rPaVZC_93  rPaVZC_94  rPaVZC_95
\
selling_period

96                                 0          1          4          0

97                                 3          1          3          0

98                                 3          3          1          0

99                                 3          0          0          1

100                                3          0          0          1


unique_selling_season_key  rPaVZC_96  rPaVZC_97  rPaVZC_98  rPaVZC_99

selling_period

96                                 0          0          1          1

97                                 3          0          0          1

98                                 1          0          0          0

99                                 4          0          0          0
```

```
100                                      1          0          0          0

[5 rows x 400 columns]
```

compute the utilization fractions

```
for c in dfp_util.columns:
    dfp_util[c] = dfp_util[c].cumsum()
    dfp_util[c] = round(dfp_util[c] / 80, 2)

dfp_util.tail(15)
```

```
unique_selling_season_key  3HmCSH_1  3HmCSH_10  3HmCSH_100  3HmCSH_11
\
selling_period

86                             0.38       0.37        0.38       0.40

87                             0.39       0.38        0.39       0.41

88                             0.40       0.39        0.41       0.42

89                             0.41       0.40        0.42       0.44

90                             0.43       0.42        0.43       0.45

91                             0.44       0.43        0.44       0.46

92                             0.45       0.44        0.45       0.47

93                             0.46       0.45        0.47       0.48

94                             0.48       0.46        0.48       0.49

95                             0.49       0.47        0.49       0.51

96                             0.50       0.49        0.50       0.52

97                             0.51       0.50        0.52       0.53

98                             0.53       0.51        0.53       0.54

99                             0.54       0.52        0.54       0.56

100                            0.55       0.54        0.55       0.57


unique_selling_season_key  3HmCSH_12  3HmCSH_13  3HmCSH_14  3HmCSH_15
\
selling_period
```

```
86                                0.39       0.40       0.37       0.32

87                                0.40       0.41       0.38       0.33

88                                0.41       0.42       0.39       0.34

89                                0.42       0.43       0.41       0.35

90                                0.43       0.44       0.42       0.36

91                                0.44       0.45       0.43       0.37

92                                0.45       0.46       0.44       0.38

93                                0.47       0.47       0.46       0.39

94                                0.48       0.48       0.47       0.39

95                                0.49       0.49       0.48       0.40

96                                0.50       0.51       0.49       0.41

97                                0.52       0.52       0.51       0.42

98                                0.53       0.53       0.52       0.43

99                                0.54       0.54       0.53       0.44

100                               0.55       0.55       0.54       0.45


unique_selling_season_key  3HmCSH_16  3HmCSH_17  ...  rPaVZC_90
rPaVZC_91  \
selling_period                                   ...

86                                0.33       0.38  ...       0.38
0.36
87                                0.34       0.39  ...       0.39
0.37
88                                0.35       0.40  ...       0.40
0.38
89                                0.36       0.41  ...       0.41
0.39
90                                0.37       0.42  ...       0.42
0.40
91                                0.38       0.43  ...       0.43
0.41
92                                0.39       0.45  ...       0.44
0.42
93                                0.40       0.46  ...       0.45
```

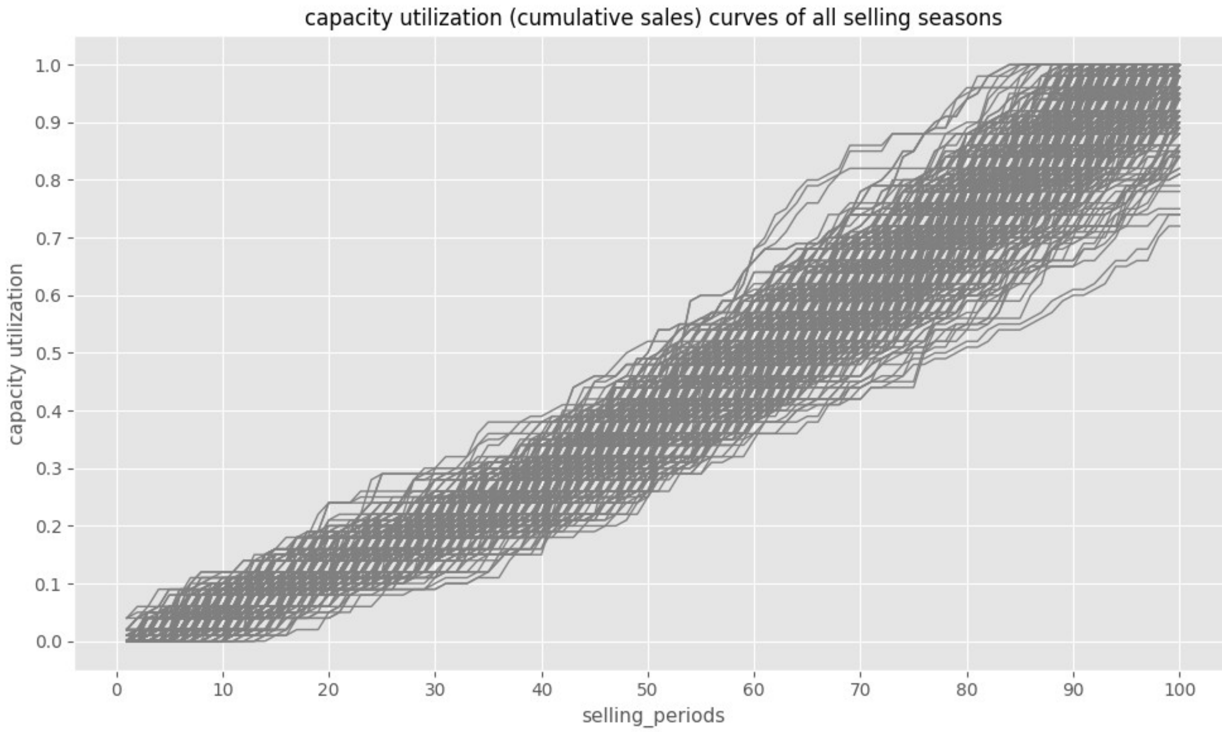
```
0.44
94                                0.41       0.47  ...       0.46
0.45
95                                0.43       0.48  ...       0.47
0.46
96                                0.44       0.49  ...       0.48
0.47
97                                0.45       0.51  ...       0.50
0.48
98                                0.46       0.52  ...       0.51
0.49
99                                0.47       0.53  ...       0.52
0.51
100                               0.49       0.54  ...       0.53
0.52

unique_selling_season_key  rPaVZC_92  rPaVZC_93  rPaVZC_94  rPaVZC_95
\
selling_period

86                              0.36       0.35       0.31       0.38

87                              0.37       0.36       0.32       0.39

88                              0.38       0.37       0.33       0.40

89                              0.39       0.38       0.34       0.41

90                              0.40       0.39       0.35       0.42

91                              0.41       0.40       0.36       0.43

92                              0.43       0.41       0.37       0.44

93                              0.44       0.42       0.38       0.46

94                              0.45       0.43       0.40       0.47

95                              0.46       0.44       0.41       0.48

96                              0.47       0.45       0.42       0.49

97                              0.48       0.46       0.43       0.51

98                              0.49       0.47       0.44       0.52

99                              0.50       0.48       0.46       0.53

100                             0.51       0.49       0.47       0.54
```

unique_selling_season_key	rPaVZC_96	rPaVZC_97	rPaVZC_98	rPaVZC_99
selling_period				
86	0.33	0.40	0.38	0.38
87	0.34	0.41	0.39	0.39
88	0.34	0.42	0.40	0.40
89	0.35	0.43	0.42	0.41
90	0.36	0.45	0.43	0.42
91	0.37	0.46	0.44	0.43
92	0.38	0.47	0.45	0.45
93	0.39	0.48	0.46	0.46
94	0.40	0.49	0.48	0.47
95	0.41	0.51	0.49	0.48
96	0.42	0.52	0.50	0.49
97	0.43	0.53	0.51	0.51
98	0.44	0.54	0.53	0.52
99	0.45	0.56	0.54	0.53
100	0.46	0.57	0.55	0.54

[15 rows x 400 columns]

```
fig, ax = plt.subplots(figsize=(10,6))

ax.plot(dfp_util.index, dfp_util, color='grey',
        linestyle='-', linewidth=1, )
ax.set_xlabel('selling_periods', size=11)
# ax.set_xticks(np.arange(min(dft.index), max(dft.index)+1, 1.0))
ax.set_ylabel("capacity utilization" , size=11)

y_majorLocator = MultipleLocator(0.1)
ax.yaxis.set_major_locator(y_majorLocator)
x_majorLocator = MultipleLocator(10)
ax.xaxis.set_major_locator(x_majorLocator)

ax.set_title("capacity utilization (cumulative sales) curves of all
```

```
selling seasons", size=12)
# ax.legend(loc=0)

ax.grid(True)
fig.tight_layout()

plt.show()
```



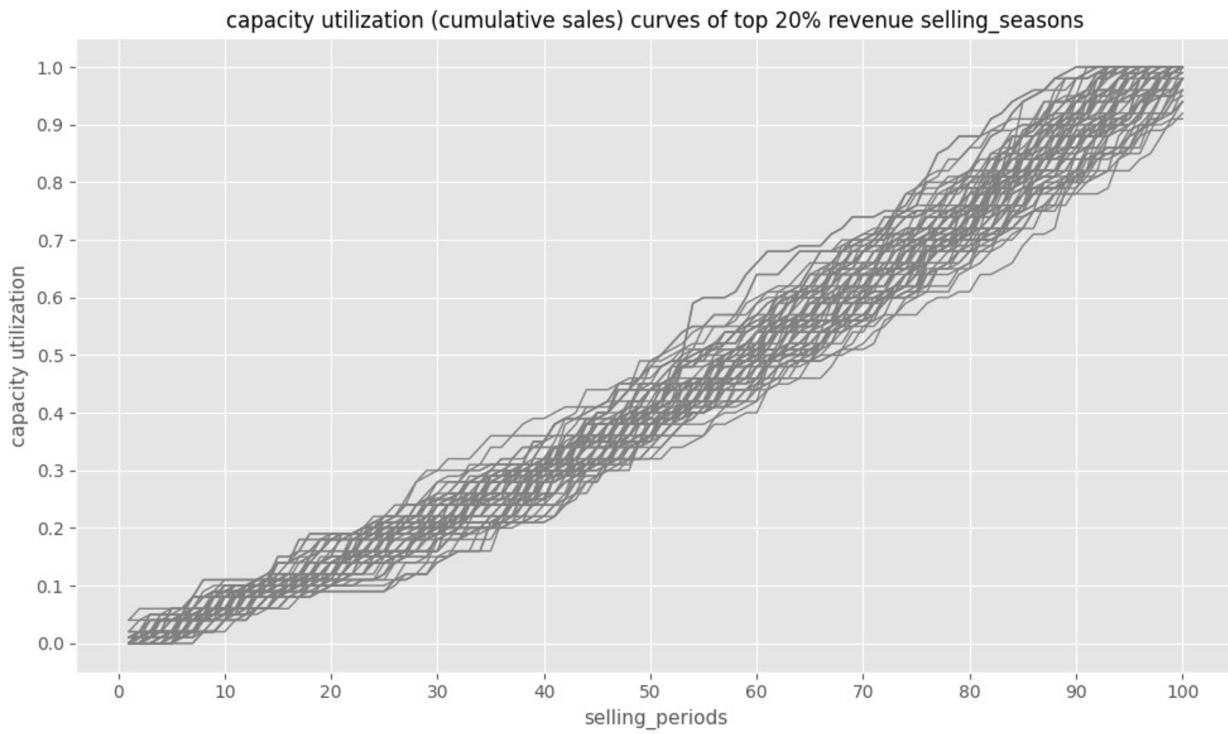
show only the top revenue curves

```
dfp_top_curves = dfp_util[   """ADD CODE HERE"""   ]
dfp_top_curves.shape
```

```
(100, 80)
```

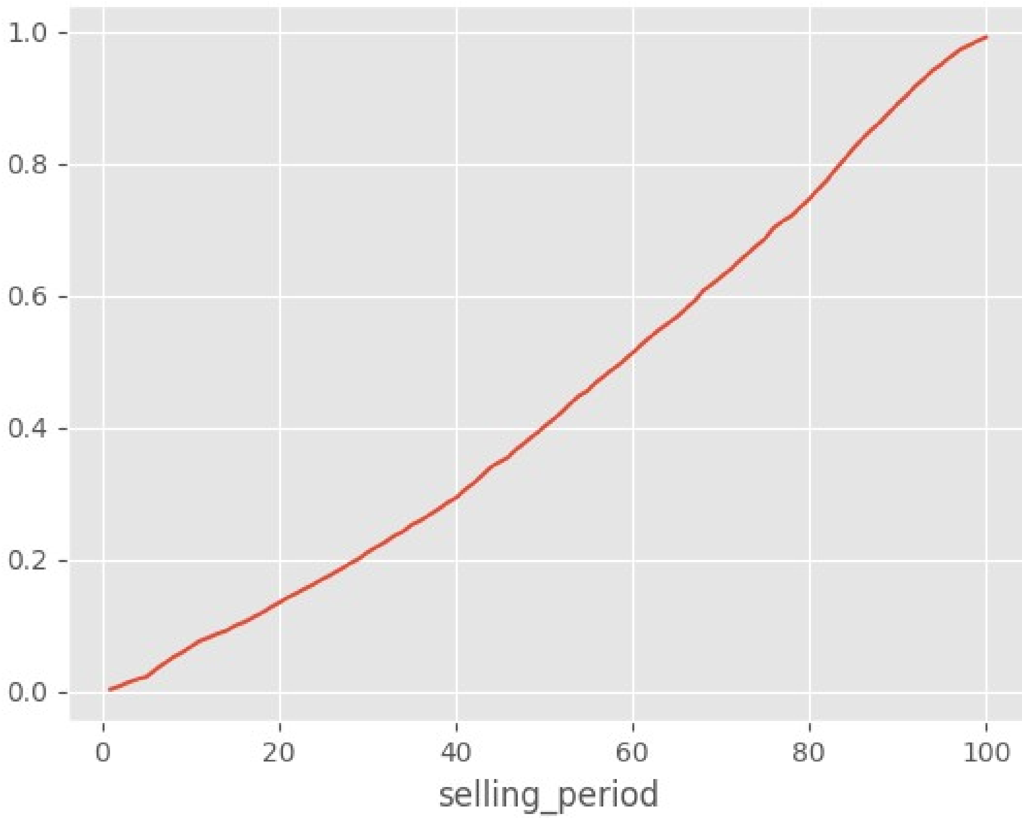
```
fig, ax = plt.subplots(figsize=(10,6))

ax.plot(dfp_top_curves.index, dfp_top_curves, color='grey',
        linestyle='-', linewidth=1, )
ax.set_xlabel('selling_periods', size=11)
# ax.set_xticks(np.arange(min(dft.index), max(dft.index)+1, 1.0))
ax.set_ylabel("capacity utilization" , size=11)

y_majorLocator = MultipleLocator(0.1)
ax.yaxis.set_major_locator(y_majorLocator)
x_majorLocator = MultipleLocator(10)
ax.xaxis.set_major_locator(x_majorLocator)
```

```
ax.set_title("capacity utilization (cumulative sales) curves of top 20% revenue selling_seasons", size=12)
# ax.legend(loc=0)

ax.grid(True)
fig.tight_layout()

plt.show()
```



get the mean curve of the top revenue generating ones

```
dfp_top_curves['mean_curve'] = """ADD CODE HERE"""
dfp_top_curves['mean_curve'].plot()

<Axes: xlabel='selling_period'>
```




```
dfp_top_curves

unique_selling_season_key  3UDGYU_57  3UDGYU_96  bh2y6f_1
bh2y6f_11  \
selling_period

1                               0.01       0.00       0.00       0.00

2                               0.01       0.00       0.00       0.00

3                               0.01       0.00       0.00       0.00

4                               0.02       0.00       0.01       0.00

5                               0.02       0.00       0.01       0.00

...                              ...        ...        ...        ...

96                              1.00       0.98       0.92       0.98

97                              1.00       0.98       0.96       1.00

98                              1.00       0.98       0.99       1.00

99                              1.00       0.99       1.00       1.00
```

```
100                                1.00       1.00      1.00       1.00

unique_selling_season_key  bh2y6f_14  bh2y6f_15  bh2y6f_19  bh2y6f_21
\
selling_period

1                                0.0       0.00       0.00       0.00

2                                0.0       0.01       0.01       0.00

3                                0.0       0.02       0.02       0.01

4                                0.0       0.04       0.02       0.01

5                                0.0       0.04       0.02       0.01

...                              ...        ...        ...        ...

96                               1.0       0.98       1.00       1.00

97                               1.0       0.99       1.00       1.00

98                               1.0       0.99       1.00       1.00

99                               1.0       1.00       1.00       1.00

100                              1.0       1.00       1.00       1.00

unique_selling_season_key  bh2y6f_22  bh2y6f_23  ...  rPaVZC_80
rPaVZC_84  \
selling_period                                   ...

1                               0.00       0.00  ...       0.00
0.01
2                               0.00       0.00  ...       0.00
0.01
3                               0.01       0.00  ...       0.00
0.01
4                               0.01       0.00  ...       0.01
0.01
5                               0.02       0.00  ...       0.01
0.01
...                              ...        ...  ...        ...
...
96                              0.88       0.90  ...       0.94
0.88
97                              0.89       0.92  ...       0.95
0.92
```

```
98                                0.89       0.92  ...       0.98
0.92
99                                0.90       0.92  ...       0.99
0.92
100                               0.92       0.96  ...       1.00
0.94

unique_selling_season_key  rPaVZC_90  rPaVZC_91  rPaVZC_92  rPaVZC_95
\
selling_period

1                               0.00       0.02       0.01       0.00

2                               0.00       0.02       0.02       0.00

3                               0.01       0.02       0.05       0.00

4                               0.04       0.04       0.05       0.00

5                               0.04       0.04       0.05       0.01

...                              ...        ...        ...        ...

96                              0.96       0.92       0.85       0.98

97                              0.98       0.95       0.89       0.98

98                              0.99       0.98       0.92       0.98

99                              1.00       1.00       0.96       0.99

100                             1.00       1.00       1.00       1.00


unique_selling_season_key  rPaVZC_97  rPaVZC_98  rPaVZC_99  mean_curve

selling_period

1                               0.00       0.00       0.00    0.003125

2                               0.01       0.01       0.01    0.007750

3                               0.01       0.02       0.04    0.013375

4                               0.01       0.02       0.04    0.018375

5                               0.02       0.02       0.04    0.021750

...                              ...        ...        ...         ...

96                              1.00       1.00       0.99    0.961750
```

```
97                                    1.00        1.00        1.00    0.971875

98                                    1.00        1.00        1.00    0.978500

99                                    1.00        1.00        1.00    0.985375

100                                   1.00        1.00        1.00    0.991500


[100 rows x 81 columns]
```

generate dict to describe the cruve at every 5th selling period in the selling season

```
target_sales_curve = {}
for t in  """ADD CODE HERE""" :
    target = {}
    target['cap_util'] = round(dfp_top_curves.loc[t,'mean_curve'],2)
    target['sold_seats'] =  """ADD CODE HERE"""
    target_sales_curve[t] = target

target_sales_curve

{1: {'cap_util': 0.0, 'sold_seats': 0},
 6: {'cap_util': 0.03, 'sold_seats': 3},
 11: {'cap_util': 0.08, 'sold_seats': 6},
 16: {'cap_util': 0.1, 'sold_seats': 8},
 21: {'cap_util': 0.14, 'sold_seats': 11},
 26: {'cap_util': 0.18, 'sold_seats': 14},
 31: {'cap_util': 0.22, 'sold_seats': 17},
 36: {'cap_util': 0.26, 'sold_seats': 21},
 41: {'cap_util': 0.3, 'sold_seats': 24},
 46: {'cap_util': 0.36, 'sold_seats': 28},
 51: {'cap_util': 0.41, 'sold_seats': 33},
 56: {'cap_util': 0.47, 'sold_seats': 38},
 61: {'cap_util': 0.52, 'sold_seats': 42},
 66: {'cap_util': 0.58, 'sold_seats': 46},
 71: {'cap_util': 0.64, 'sold_seats': 51},
 76: {'cap_util': 0.7, 'sold_seats': 56},
 81: {'cap_util': 0.76, 'sold_seats': 61},
 86: {'cap_util': 0.84, 'sold_seats': 67},
 91: {'cap_util': 0.9, 'sold_seats': 72},
 96: {'cap_util': 0.96, 'sold_seats': 77}}

with open('target_sales_curve.pkl', "wb") as f:
    pickle.dump(target_sales_curve, f, 
protocol=pickle.HIGHEST_PROTOCOL)
```

(2) implement the logic to follow this capacity utilization curve by controlling the price to stay close to the curve

price up -> slow down sales

price down -> increase sales

prepare given demand input for given competition and given selling season

```
with open('target_sales_curve.pkl', 'rb') as f:
            target_curve = pickle.load(f)
target_curve
```

```
{1: {'cap_util': 0.0, 'sold_seats': 0},
 6: {'cap_util': 0.03, 'sold_seats': 3},
 11: {'cap_util': 0.08, 'sold_seats': 6},
 16: {'cap_util': 0.1, 'sold_seats': 8},
 21: {'cap_util': 0.14, 'sold_seats': 11},
 26: {'cap_util': 0.18, 'sold_seats': 14},
 31: {'cap_util': 0.22, 'sold_seats': 17},
 36: {'cap_util': 0.26, 'sold_seats': 21},
 41: {'cap_util': 0.3, 'sold_seats': 24},
 46: {'cap_util': 0.36, 'sold_seats': 28},
 51: {'cap_util': 0.41, 'sold_seats': 33},
 56: {'cap_util': 0.47, 'sold_seats': 38},
 61: {'cap_util': 0.52, 'sold_seats': 42},
 66: {'cap_util': 0.58, 'sold_seats': 46},
 71: {'cap_util': 0.64, 'sold_seats': 51},
 76: {'cap_util': 0.7, 'sold_seats': 56},
 81: {'cap_util': 0.76, 'sold_seats': 61},
 86: {'cap_util': 0.84, 'sold_seats': 67},
 91: {'cap_util': 0.9, 'sold_seats': 72},
 96: {'cap_util': 0.96, 'sold_seats': 77}}
```

load varriables as given in the duopoly p() function from our input data

```
# take first selling season
id = df_comp_details['unique_selling_season_key'].unique()[0]
df_select =
df_comp_details[df_comp_details.unique_selling_season_key==id]

# define selling period
selling_period_in_current_season = 31

# get according deman object
demand_historical_in_current_season  = np.array(
            df_select[df_select.selling_period <
selling_period_in_current_season]['demand']
```

```
)
demand_historical_in_current_season

array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 2, 0, 2, 1, 1, 2,
0,
       0, 0, 0, 0, 0, 0, 1, 0])

# get according price object
prices_historical_in_current_season = [
    df_select[df_select.selling_period <
selling_period_in_current_season]['price'],
    df_select[df_select.selling_period <
selling_period_in_current_season]['price_competitor'] ]
prices_historical_in_current_season  =
np.array(prices_historical_in_current_season)

prices_historical_in_current_season

array([[66.4, 66.4, 66.4, 66.4, 66.4, 33.2, 33.2, 33.2, 33.2, 33.2,
16.6,
        16.6, 16.6, 16.6, 16.6, 11.9, 11.9, 11.9, 11.9, 11.9, 10.7,
10.7,
        10.7, 10.7, 10.7,  9. ,  9. ,  9. ,  9. ,  9. ],
       [56.1, 66.4, 66.4, 66.4, 66.4, 66.4, 33.2, 33.2, 33.2, 33.2,
33.2,
        16.6, 16.6, 16.6, 16.6, 16.6, 11.9, 11.9, 11.9, 11.9, 11.9,
10.7,
        10.7, 10.7, 10.7, 10.7,  9. ,  9. ,  9. ,  9. ]])
```

logic

```
last_price = prices_historical_in_current_season[0][-1]
last_price

9.0

current_sold_seats = demand_historical_in_current_season.sum()
current_sold_seats

12

print("old price: %.2f" % last_price)

if selling_period_in_current_season in target_curve.keys():

    target = target_curve[selling_period_in_current_season]
    target_sales = target['sold_seats']

    print("sold seats: %d " % current_sold_seats)
    print("target sold seats: %d " % target_sales)
```

```python
    delta_in_sales =  """ADD CODE HERE"""
    print("percentage delta to target: %.2f" % delta_in_sales)

    ###
    # adjust price according to the delta
    price =  """ADD CODE HERE"""
    print("new price: %.2f" % price)
```

```
old price: 9.00
sold seats: 12
target sold seats: 17
percentage delta to target: -0.29
new price: 20.00
```